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Of the many wireless technologies mushrooming around us, GPRS may be the most important for wireless communications

From cordless phones to mobile phones, wireless communication has seeped transparently into our daily lives. Wireless technologies have far reaching applications but have gained popularity and acceptance only in the field of voice communication. Wireless data access such as sending memos, exchanging e-mail, scheduling appointments, etc is still not widespread, largely due to very slow connection speeds. Even the development of the Wireless Application Protocol (WAP) hasn't helped much—you cannot browse existing Web sites or access your data until they are converted to the WAP format.

General Packet Radio Service (GPRS) is the first wireless service that addresses the need to access any information at all times. It provides an always-on connection to the Internet and works with existing Internet standards, tremendously increasing its applications. GPRS works over the existing Global System for Mobile Communications (GSM) networks, making it easy and inexpensive to set up, since this is used by most mobile service providers today.

Other mobile service providers use a standard called Time Division Multiple Access (TDMA). They are also tilting towards GPRS with a wireless technology called Enhanced Data Rates for Global Evolution (EDGE). EDGE and GPRS are both based on similar infrastructure and protocols, except that EDGE uses a more sophisticated radio interface.

The family tree

Cellular technologies have come a long way since their birth, evolving in phases called Generations (G). Each generation takes the technology a step further, providing exponentially higher data rates and better bandwidth efficiency, and allows the possibility of more mobile applications.

Second generation (2G) protocols include GSM, TDMA and CDMA (Code Division Multiple Access), and use digital encoding. The 2G network protocols support voice and some limited data